

Project Title : Home Automation System
Course Code : EEE 4604
Course Title : Measurement and Instrumentation Lab
Project Mark : 30 Marks

Description:

The objective of this project is to design and implement a home automation system utilizing different sensors to achieve specific automated functionalities. The system should be fully implemented in hardware and accompanied by a comprehensive project report.

Objectives:

System Activation via LDR Sensor:

The system will be initiated using a Light Dependent Resistor (LDR) sensor. A physical object (any obstacle like paper card), which does not allow light to pass through (an optical obstacle), will be placed in front of the LDR. Upon detecting the absence of the light, the system will be activated, and all subsequent sensors will become operational.

Temperature Monitoring and Motor Activation:

A temperature sensor will be incorporated to monitor ambient temperature levels. If the measured temperature exceeds a predefined threshold, a motor will be automatically activated.

Smoke Detection and Alert:

A smoke detector will be included as a safety measure. Upon detecting smoke, the system will trigger an audible alert using a buzzer. A manual resetting option will be available to turn off the buzzer after the activation of the buzzer.

Water Level Monitoring:

An ultrasonic sensor will be used to monitor the water level within a water tank. The current water level must be displayed in real time using a digital display interface.

Automatic Water Dispensing:

An infrared (IR) sensor will be used to detect the presence of an object (such as a hand) near a water tap. When an object is detected, water will be dispensed from the tap for as long as the object remains in front of the sensor. The water must be supplied from the previously installed tank within the system.

Requirements:

- Hardware Prototype
- Project Report
 - Project Title
 - Group Name
 - Group-mate names and IDs
 - Introduction of the project
 - Components list and their properties
 - Working Principle of the entire project
 - Flowchart representation of the working principle
 - Wiring Diagram / Project Picture
 - Code (if any)
 - Cost Breakdown
 - Limitations / Future Scopes
 - Conclusion

Project Submission Deadline: 5th September 2025

Mark Distribution:

Sl. No.	Phase	Marks
1.	Implementation of the project through Hardware	20
2.	Project Presentation	05
3.	Project Report	05